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# Infrastructure is Dead, Long Live Infrastructure in the Age of Agentic AI

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PRESENTATION OVERVIEW

# Infrastructure is Dead,
# Long Live Infrastructure in the Age of Agentic AI

The rise of Agentic AI is fundamentally transforming how enterprise Azure environments are architected, designed, deployed, and managed. Traditional infrastructure practices — once reliant on manual processes and static configurations — are giving way to intelligent, autonomous systems capable of making real-time decisions at scale. By integrating Agentic AI with DevOps pipelines, organisations can achieve self-optimising architectures, accelerated deployments, and continuous operational intelligence. This presentation explores how this convergence redefines the infrastructure lifecycle — shifting engineers from reactive administrators to strategic orchestrators in an increasingly autonomous, AI-driven cloud landscape.

Agentic AI | Azure | DevOps | Enterprise Cloud | Infrastructure

AGENDA

# Session Overview

## Infrastructure is Dead — Long Live Infrastructure in the Age of Agentic AI

**Open (0–8 min) Setting the scene + opening poll**

What Agentic AI actually means, and where your org sits today

**Part 1 (8–22 min) The infrastructure paradigm shift**

From manual pipelines to autonomous, intent-driven architectures

**Part 2 (22–38 min) Agents in the DevOps lifecycle**

Pipeline integration, live demo, guardrails and governance + Q&A

**Part 3 (38–52 min) Real-world application**

Two case studies, audience poll, and addressing your concerns

**Part 4 (52–60 min) Getting started + open Q&A**

The engineer's new role, adoption roadmap, key takeaways

Agentic AI | Azure | DevOps | Enterprise Cloud | Infrastructure

OPENING POLL

# Where does your organisation sit on the cloud automation maturity curve?

**Raise your hand for the option that best describes you today:**

**Level 1 — Manual**

Infrastructure provisioned by hand; runbooks and tickets rule the day

**Level 2 — Scripted**

Terraform or Bicep in place, but humans drive every pipeline trigger

**Level 3 — Automated**

CI/CD pipelines running reliably; some self-healing in production

**Level 4 — Intelligent**

AI-assisted code review, anomaly detection, or auto-remediation in play

Poll | Audience Participation | Cloud Maturity

FOUNDATIONS

# What is Agentic AI?

## And what it definitely is not

### A working definition

An Agentic AI system perceives its environment, reasons over a goal, plans a sequence of actions, executes those actions using tools, and adapts based on results — with minimal human intervention at each step.

### Not a chatbot

Copilots suggest. Agents act. The difference is autonomy, memory, and tool use.

### Not pure automation

Scripts follow fixed rules. Agents reason about context and handle novel situations.

### Key capabilities that make it work

Goal decomposition, tool calling, memory, multi-step planning, and feedback loops

Agentic AI | Foundations | LLM | Azure AI Foundry

PART 1

# The Old World

## Manual pipelines, static configs, reactive operations

**How infrastructure worked (and in many places still does)**

**Ticket-driven provisioning**

Engineers open requests; wait times measured in hours or days; human bottlenecks at every stage

**Static configuration drift**

IaC templates describe desired state but nothing enforces it continuously; drift accumulates silently

**Reactive incident response**

Alerts fire, engineers wake up, they dig through logs, identify root cause, apply fix — repeat

**Knowledge locked in individuals**

Runbooks age out, tribal knowledge walks out the door, onboarding takes months

Legacy Ops | Manual Provisioning | Incident Response | IaC

PART 1

# The New World

## Autonomous agents, intent-driven infrastructure, proactive operations

**What the AI-native infrastructure paradigm looks like**

**Intent-driven provisioning**

An engineer describes the outcome in natural language; an agent reasons, generates IaC, validates, and deploys

**Continuous compliance enforcement**

Agents monitor configuration state in real time and remediate drift before it causes incidents

**Proactive self-healing**

Agents detect anomalies, correlate signals across services, and remediate — often before an alert fires

**Institutional knowledge encoded**

Runbooks become agent prompts; best practices are embedded in guardrails; expertise scales across teams

Agentic AI | Self-Healing | Intent-Driven | Azure

PART 1

# Agentic AI Architecture on Azure

## The building blocks available to you today

**Azure AI Foundry**

Model deployment, orchestration, and evaluation — the core runtime for your agents

**Semantic Kernel**

Microsoft's open-source SDK for building agent workflows that call Azure services as tools

**GitHub Copilot for Azure**

AI-assisted IaC generation, PR reviews, and pipeline troubleshooting inside your existing workflow

**Azure DevOps + Pipelines**

The CI/CD backbone that agents plug into to trigger, monitor, and validate deployments

**Azure Policy + Defender for Cloud**

The guardrail layer — agents can query, enforce, and remediate against policy definitions

Azure AI Foundry | Semantic Kernel | GitHub Copilot | Azure DevOps

PART 1

# How Agents Interact with Azure

## ARM, Bicep, Terraform, REST APIs — the agent toolbox

**Tool calling as the integration mechanism**

Agents don't have direct access to Azure — they call functions you expose. Each function wraps an Azure operation.

**Infrastructure as code generation**

Agent reasons about requirements, generates Bicep or Terraform, validates with what-if/plan, then executes via pipeline

**ARM and REST API calls**

For dynamic operations (scaling, restarting, tagging), agents call ARM REST endpoints directly with managed identity

**Authentication and least-privilege**

Agents authenticate as managed identities with scoped RBAC — no hardcoded credentials, full audit trail in Entra ID

ARM | Bicep | Terraform | REST API | Managed Identity

PART 2

# Agents in the DevOps Lifecycle

## Plan — Provision — Monitor — Remediate

**Plan**

Agent analyses requirements, estimates cost with Azure Pricing API, flags risks, proposes architecture

**Provision**

Agent generates IaC, runs validation (what-if/plan), awaits approval gate if configured, then triggers pipeline

**Monitor**

Agent continuously queries Azure Monitor, Log Analytics, and Cost Management; surfaces anomalies with context

**Remediate**

On detecting drift or incident, agent executes a known remediation or escalates to human with full context attached

DevOps | Azure Pipelines | Azure Monitor | Remediation

LIVE DEMO

# Agentic Provisioning

Spinning up an Azure environment from a single natural language prompt

LIVE DEMO

# Demo Architecture

## Claude + GitHub Codespaces + Azure Environment

### CLAUDE

AI Agent / claude.ai

### GITHUB

Repo

Codespaces

**Preferred Scripting Language**
(Bicep / Terraform / PowerShell)

### AZURE ENVIRONMENT

**Resource Group**
(RG)

① Setup base Repo

② Setup base Azure

③ Setup base Claude

④ Vibe Code ↺ Repeat

PART 2

# Guardrails and Governance

## Policy enforcement, approval gates, and audit trails

### The trust problem — and how to solve it

Autonomous agents are only as trustworthy as the boundaries you put around them. Governance is not optional.

### Azure Policy as the rulebook

Define allowed regions, SKUs, required tags, and network configurations. Agents check policy before acting.

### Human-in-the-loop approval gates

For high-blast-radius actions (prod deployments, security group changes), agents pause and request explicit approval

### Full audit trail in Entra ID and Monitor

Every agent action is logged with the managed identity, timestamp, and operation — indistinguishable from human audit trails

Azure Policy | Governance | Compliance | RBAC | Audit

Q&A — BREAK #1

# Questions So Far?

5 minutes — anything from Parts 1 and 2

PART 3 — CASE STUDY

# Autonomous Remediation

## Multi-region Azure environment — persistent configuration drift

### The scenario

A financial services client running workloads across Australia East, Southeast Asia, and West Europe was experiencing weekly configuration drift incidents. Manual detection averaged 4 hours; remediation added another 2.

### The agentic approach

An agent was deployed to poll Azure Resource Graph every 15 minutes, compare live state against the approved Bicep baseline, and automatically remediate low-risk drift (tags, diagnostic settings, NSG rules) — escalating high-risk changes for human approval.

Case Study | Remediation | Azure Resource Graph | Multi-Region



### The outcome

Mean time to detect dropped from 4 hours to under 2 minutes. 78% of drift events were remediated autonomously

PART 3 — CASE STUDY

# Agent-Driven Cost Optimisation

## Continuous right-sizing across a 400-VM estate

### The scenario

A retail organisation running 400+ VMs across dev, test, and production environments was overspending by an estimated 35% due to over-provisioned SKUs and orphaned resources left running outside business hours.

### The agentic approach

An agent was connected to Azure Cost Management and Monitor. It generated weekly right-sizing recommendations, automatically shut down dev/test resources outside business hours, and flagged orphaned disks and NICs for review.

Case Study | Cost Optimisation | Right-Sizing | Azure Cost Management



### The outcome

Monthly Azure spend reduced by 28% within 60 days. Zero production incidents attributable to agent actions. The

POLL #2

# What is your biggest concern about adopting agentic infrastructure?

**Raise your hand for your top concern:**

**A — Security and blast radius**

An agent with write access to production is a large attack surface if compromised

**B — Auditability and compliance**

How do I prove to auditors that agent actions meet our regulatory requirements?

**C — Trust and explainability**

I don't know why the agent made a particular decision — that makes me uncomfortable

**D — Team readiness**

We don't have the skills to build, run, or debug agentic systems yet

Poll | Audience Participation | Governance | Security

PART 3

# Addressing the Concerns

## Security, trust, blast radius, and compliance — head on

**Security and blast radius**

Scope agent identity to minimum required permissions. Separate identities per environment. Network-restrict agent egress.

**Auditability and compliance**

Every agent action via managed identity appears in Entra ID audit logs. Pair with Azure Monitor alerts for immediate visibility.

**Trust and explainability**

Require agents to produce a plan and rationale before execution. Log the reasoning chain. Implement mandatory human review for Tier 1 actions.

Security | Compliance | Trust | RBAC | Governance

**Team readiness**

Start with a single low-risk use case (e.g., dev environment provisioning). Build internal capability before expanding

PART 4

# The Engineer as Orchestrator

## From reactive administrator to strategic architect

**The role is not disappearing — it is evolving**

Engineers who thrive will be those who learn to design, direct, and govern agents — not those who resist the shift.

**Skills that matter more**

Prompt engineering for infrastructure, tool design, agent orchestration, policy definition, and outcome-based thinking

**Skills that matter less**

Remembering CLI flags, writing repetitive YAML, manually chasing alerts at 2am

**The new value-add**

Designing the guardrails, defining the goals, reviewing agent outputs, and escalating edge cases — humans set the intent, agents execute

Engineers | Orchestration | Skills | Future of Work

PART 4

# A Practical Adoption Roadmap

## Crawl — Walk — Run

**Crawl — Low-risk, high-visibility wins (Months 1–3)**

Deploy an agent for dev/test environment provisioning. Instrument with full logging. Let the team get comfortable with agent outputs.

**Walk — Expand scope with human-in-the-loop (Months 3–6)**

Add cost optimisation and drift detection agents. Introduce approval gates for production actions. Build a library of tested agent tools.

**Run — Autonomous operations with governance layer (Month 6+)**

Extend to production monitoring and remediation. Define clear escalation paths. Implement an agent health dashboard and anomaly detection on agent behaviour itself.

Adoption | Roadmap | DevOps | Governance | Azure



**Guiding principle throughout**

Start narrow, instrument everything, expand only what you trust, and always maintain a clear human override path.

KEY TAKEAWAYS

# Infrastructure is Evolving,

# Not Dying — Five things to take away today

1. **Agentic AI is an amplifier, not a replacement**

The best outcomes come from human intent + agent execution. Neither alone is sufficient.

2. **Azure has the building blocks today**

AI Foundry, Semantic Kernel, GitHub Copilot, and Azure Policy form a complete agentic stack right now.

3. **Guardrails are not optional**

Governance, least-privilege identity, and approval gates are what make autonomous systems trustworthy.

4. **Start small and instrument everything**

A single well-governed use case that builds team confidence is worth more than a grand vision that stalls.

5. **The engineer's role is expanding, not shrinking**

Orchestrating agents, designing guardrails, and encoding institutional knowledge is high-value, high-impact work.

Agentic AI | Azure | DevOps | Governance | Infrastructure

Q&A — OPEN DISCUSSION

# Questions, Challenges, Ideas?

Let's talk — no such thing as a bad question

CONTACT

# Get in Touch

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